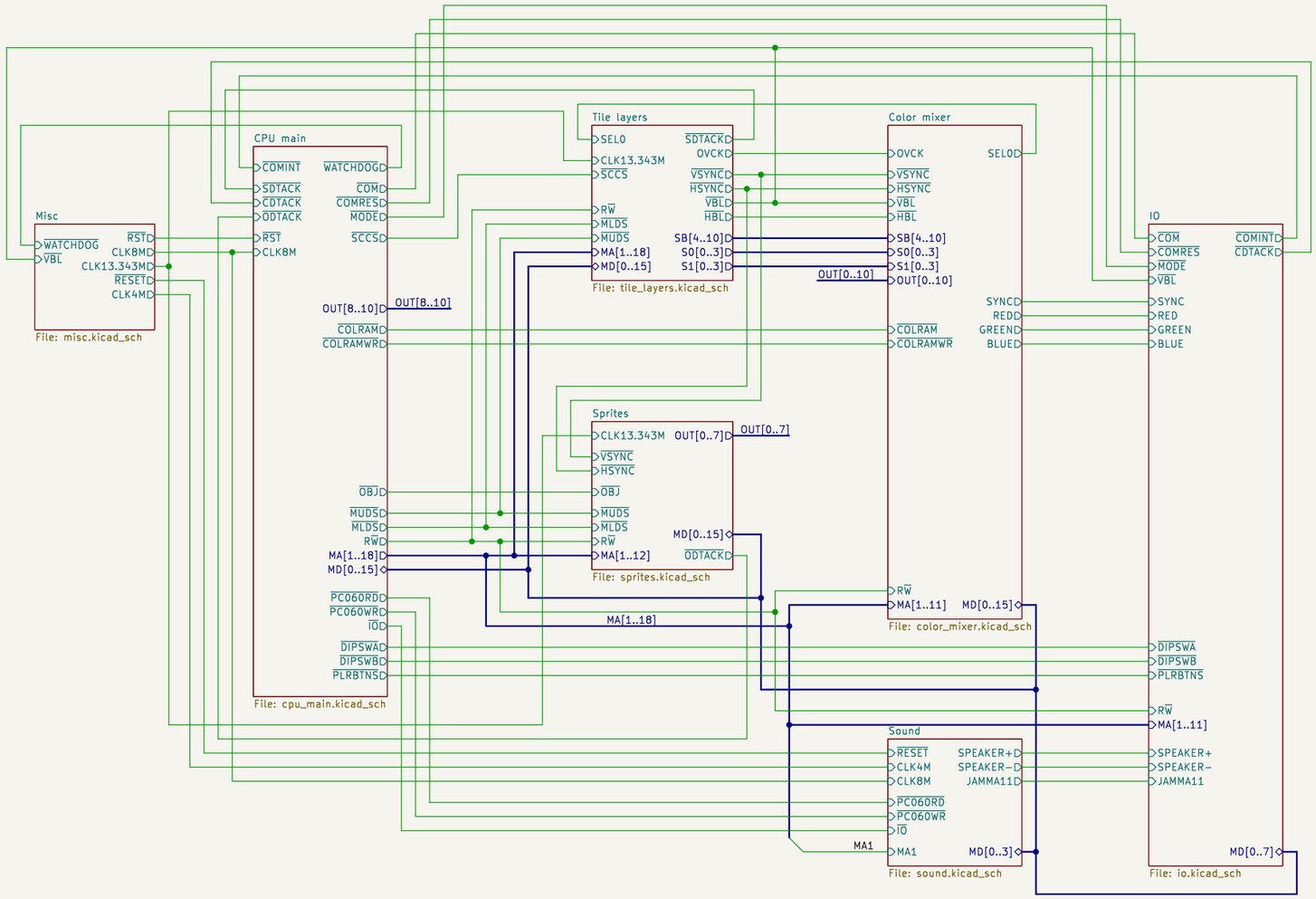


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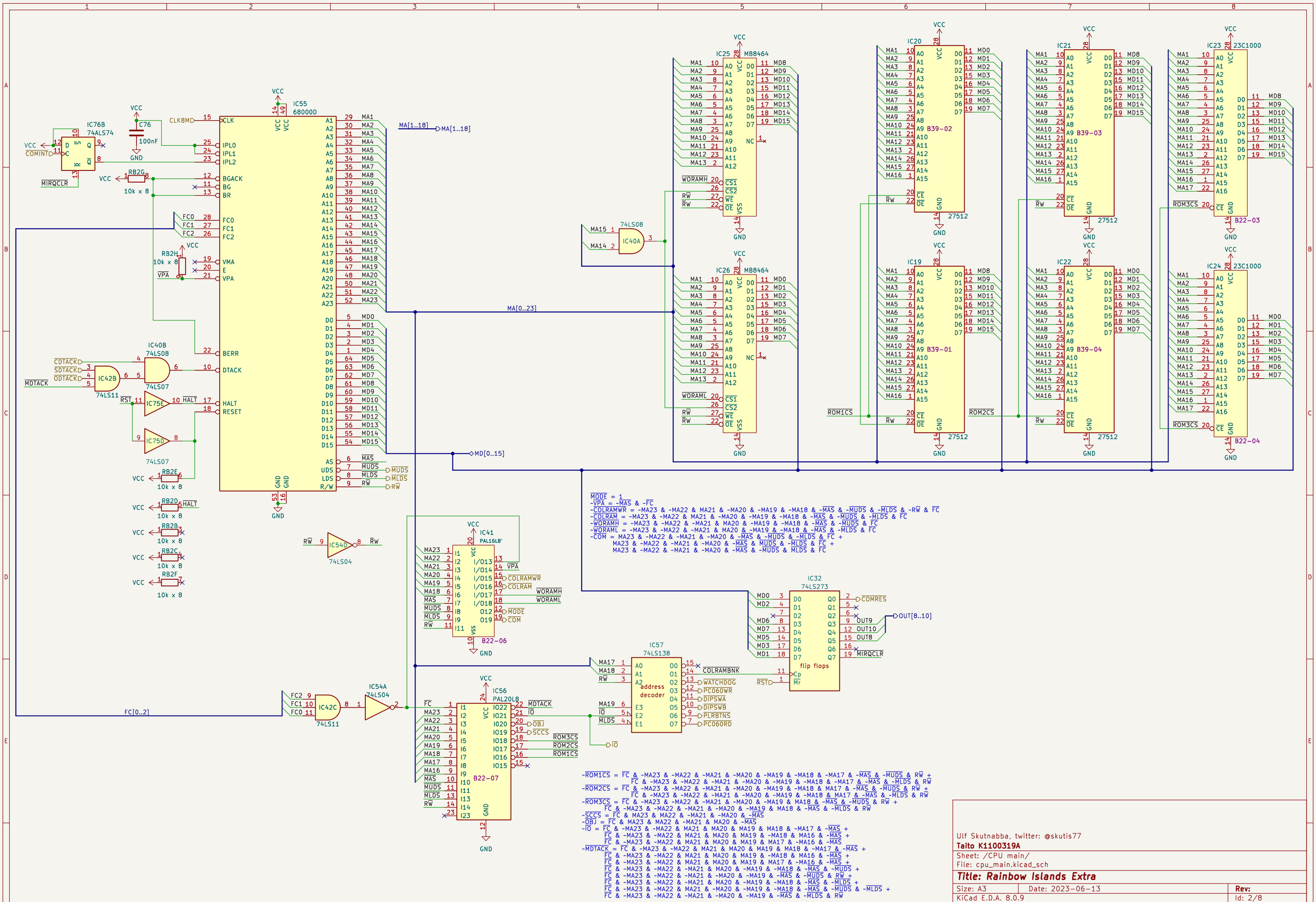
Taito K1100319A

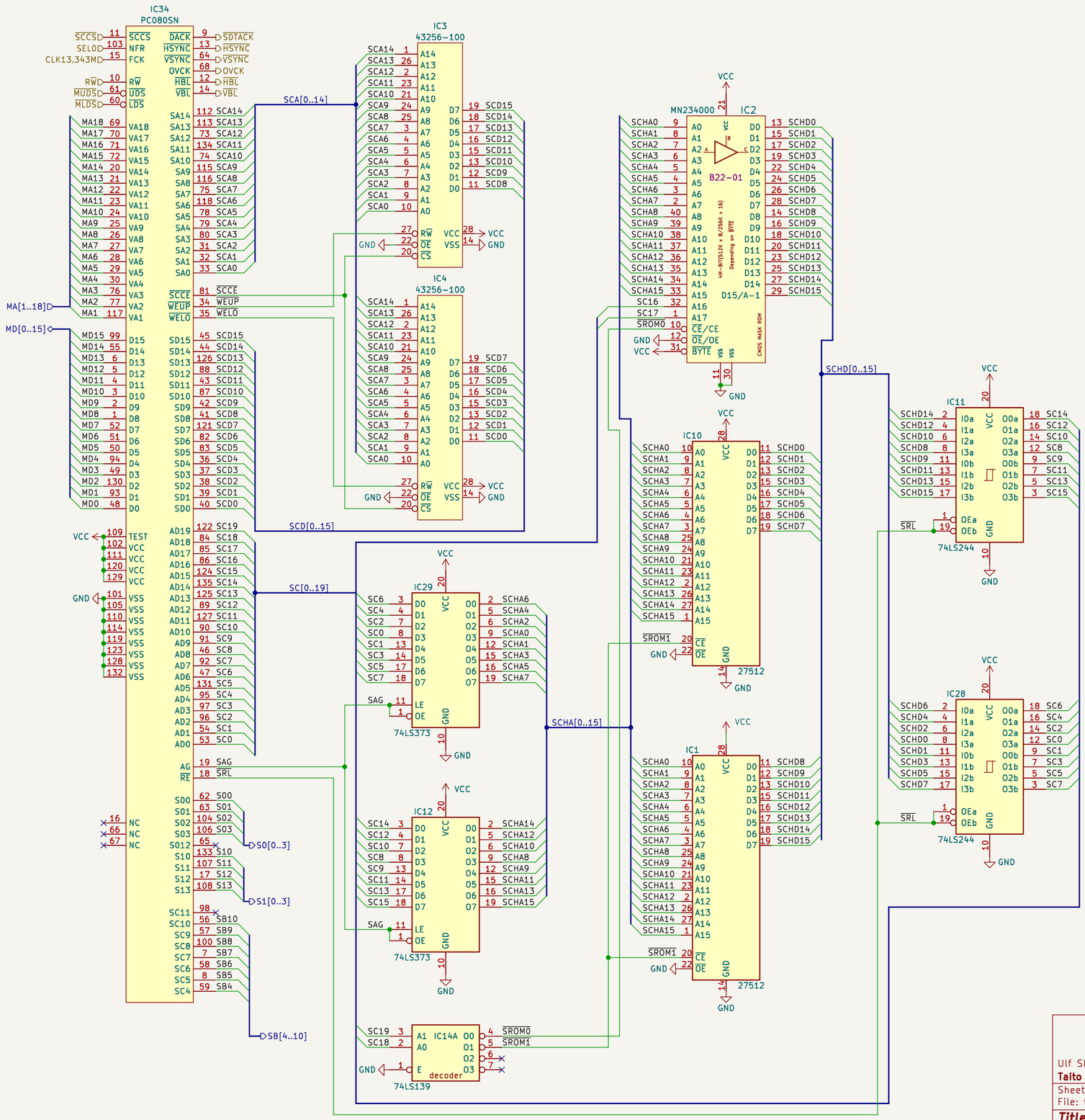
Sheet: /

File: risle.kicad_sch

Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/8


IC55
68000D
IC76B
74ALS74
C76
100nF
RB2G
10k x 8
RB2H
RB2E
RB2D
RB2B
RB2C
RB2F
IC40B
74LS08
IC42B
74LS11
IC75E
IC75D
74LS07
IC54D
74LS04
IC54A
IC42C
IC25
MB8464
IC26
IC20
IC21
IC22
IC19
27512
B39-02
B39-03
B39-04
B39-01
IC23
23C1000
IC24
B22-03
B22-04
IC40A
IC41
PAL16L8'
B22-06
IC56
PAL20L8
B22-07
IC57
74LS138
address decoder
IC32
74LS273
flip flops
MA[1..18]
MA[0..23]
MD[0..15]
FC[0..2]
OUT[8..10]
MODE = 1
-VPA = -MAS & -FC
-COLRAMWR = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & ~RW & FC
-COLRAM = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & FC
-WORAMH = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & FC
-WORAML = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS & FC
-COM = MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & ~MLDS & FC +
MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & MUDS & ~MLDS & FC +
MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & MLDS & FC
-ROM1CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MLDS & RW
-ROM2CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MLDS & RW
-ROM3CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MLDS & RW
-SCCS = FC & MA23 & MA22 & ~MA21 & ~MA20 & ~MAS
-OBJ = FC & MA23 & MA22 & ~MA21 & MA20 & ~MAS
-IO = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS
-MDTACK = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS +
FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS +
FC & ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MLDS & RW
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Taito K1100319A
Sheet: /CPU main/
File: cpu_main.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
KiCad E.D.A. 8.0.9
Rev:
Id: 2/8



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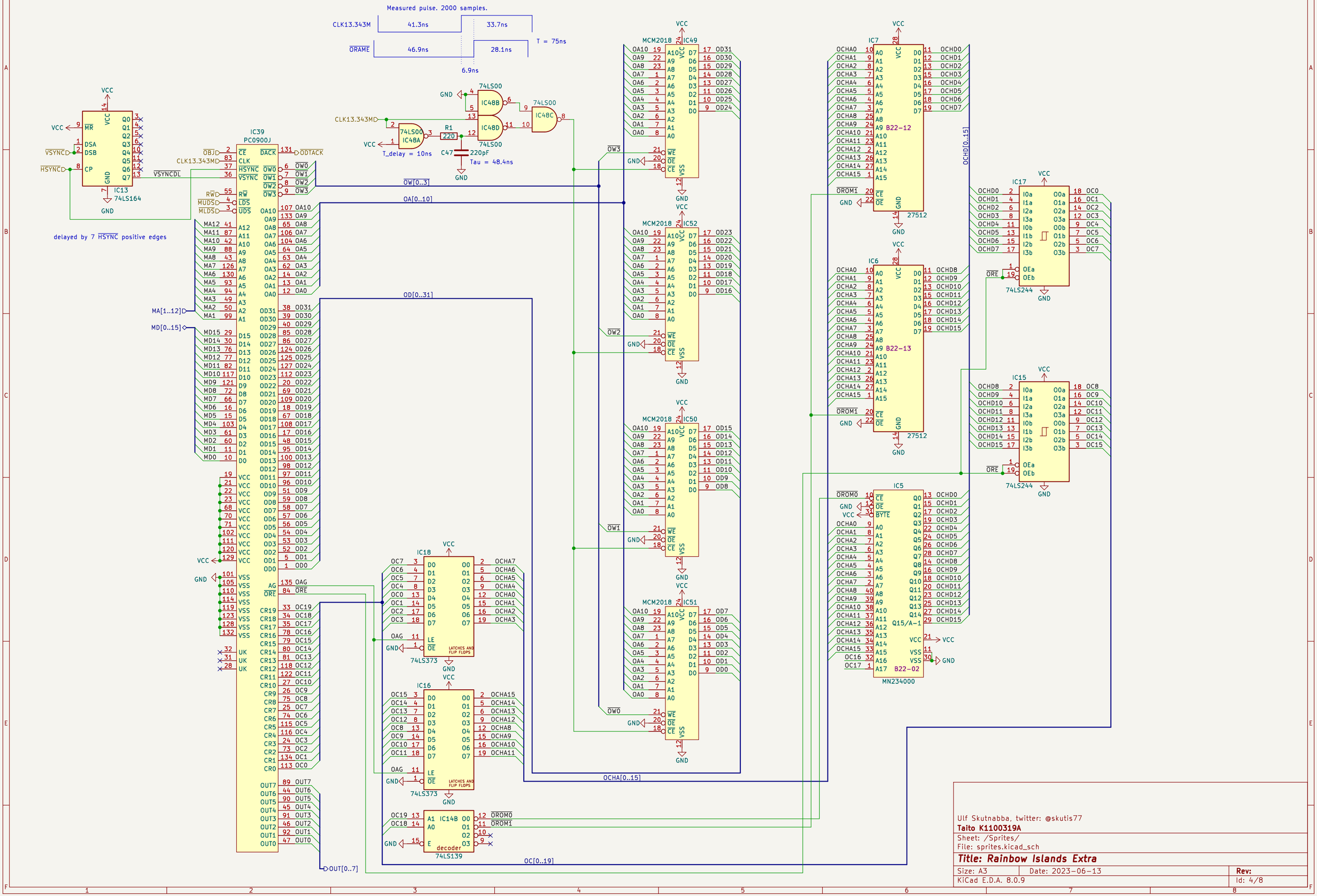
Taito K1100319A

Sheet: /Tile layers/

File: tile_layers.kicad_sch

Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
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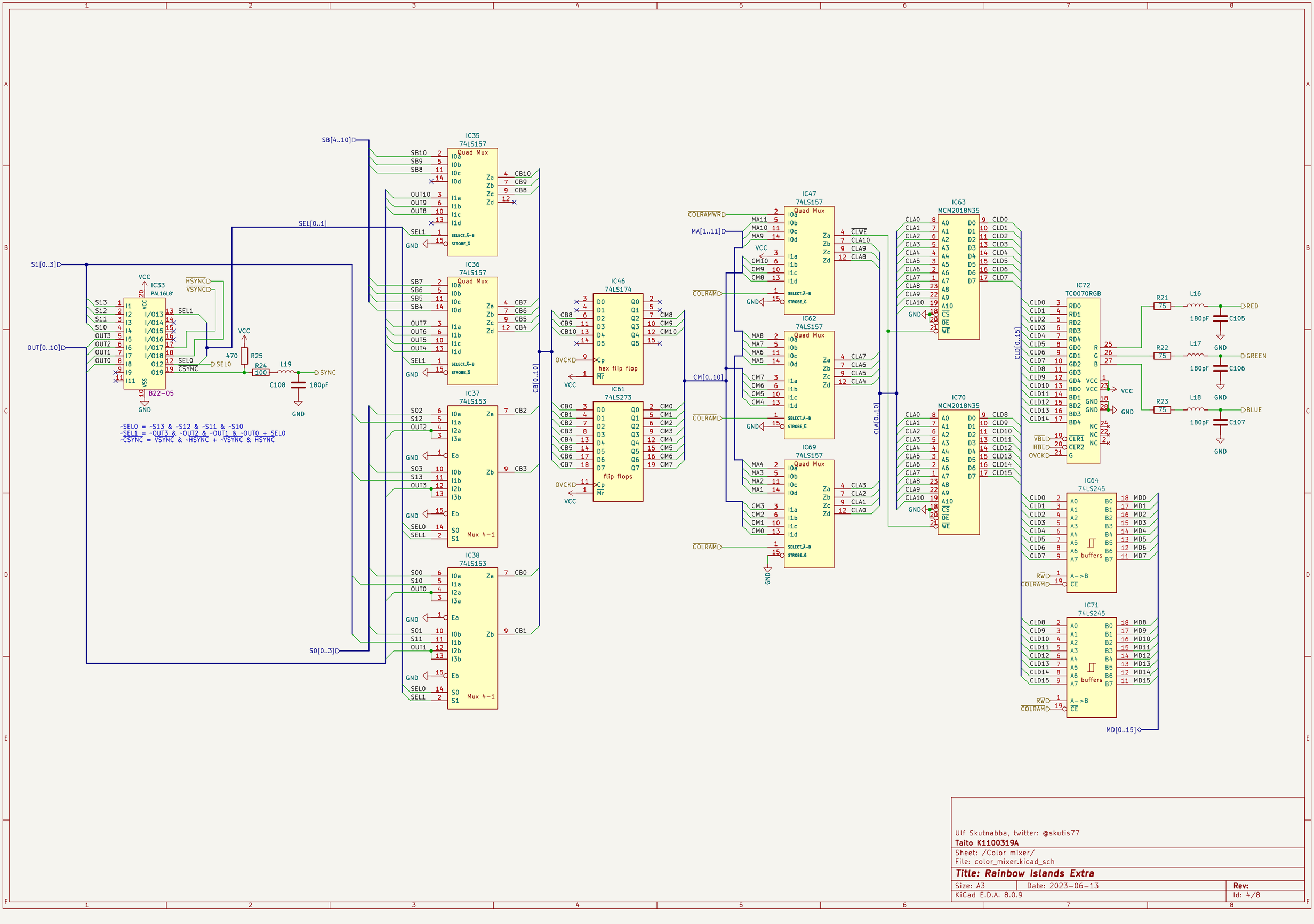
Taito K1100319A

Sheet: /Sprites/

File: sprites.kicad_sch

Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
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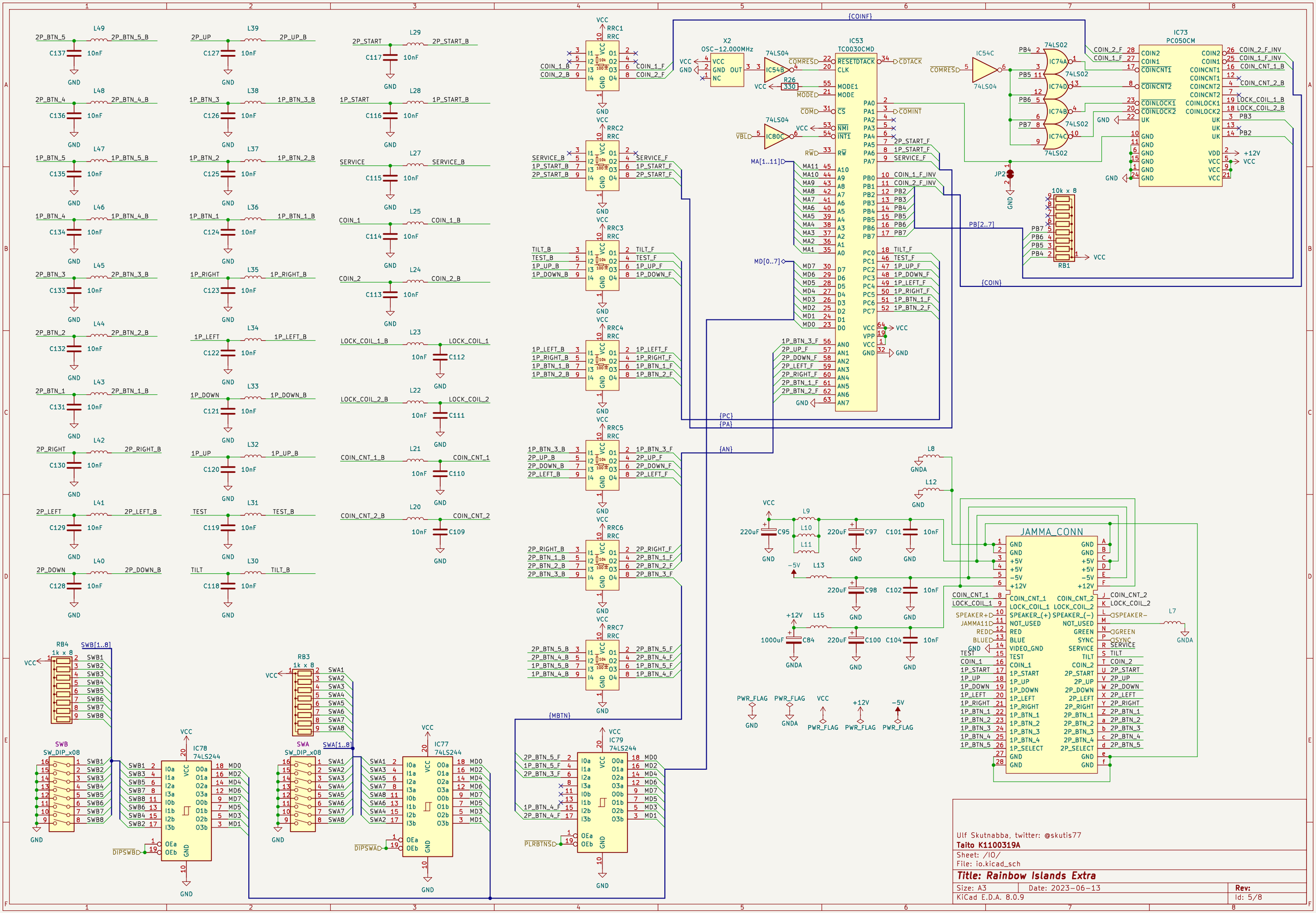
Taito K1100319A

Sheet: /Color mixer/

File: color_mixer.kicad_sch

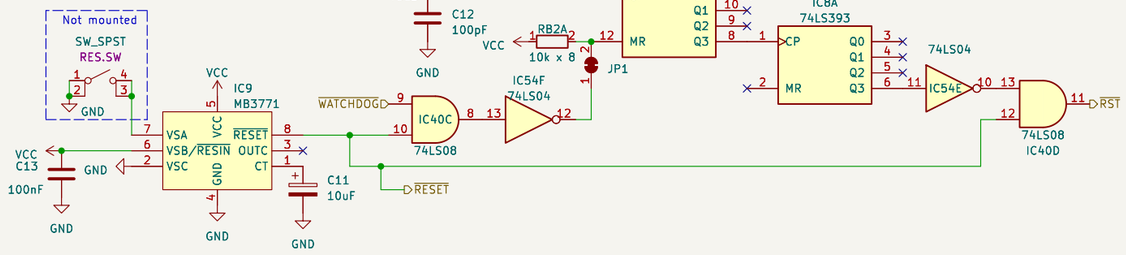
Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
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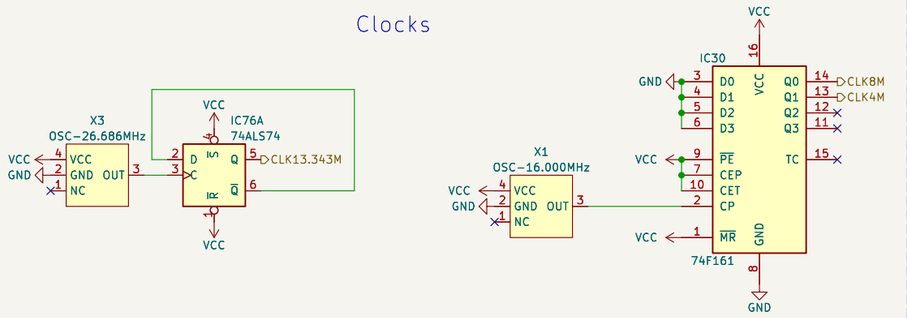

2P_BTN_5
L49
2P_BTN_5_B
C137
10nF
GND
2P_BTN_4
L48
2P_BTN_4_B
C136
1P_BTN_5
L47
1P_BTN_5_B
C135
1P_BTN_4
L46
1P_BTN_4_B
C134
2P_BTN_3
L45
2P_BTN_3_B
C133
2P_BTN_2
L44
2P_BTN_2_B
C132
2P_BTN_1
L43
2P_BTN_1_B
C131
2P_RIGHT
L42
2P_RIGHT_B
C130
2P_LEFT
L41
2P_LEFT_B
C129
2P_DOWN
L40
2P_DOWN_B
C128
2P_UP
L39
2P_UP_B
C127
1P_BTN_3
L38
1P_BTN_3_B
C126
1P_BTN_2
L37
1P_BTN_2_B
C125
1P_BTN_1
L36
1P_BTN_1_B
C124
1P_RIGHT
L35
1P_RIGHT_B
C123
1P_LEFT
L34
1P_LEFT_B
C122
1P_DOWN
L33
1P_DOWN_B
C121
1P_UP
L32
1P_UP_B
C120
TEST
L31
TEST_B
C119
TILT
L30
TILT_B
C118
2P_START
L29
2P_START_B
C117
1P_START
L28
1P_START_B
C116
SERVICE
L27
SERVICE_B
C115
COIN_1
L25
COIN_1_B
C114
COIN_2
L24
COIN_2_B
C113
LOCK_COIL_1_B
L23
LOCK_COIL_1
C112
LOCK_COIL_2_B
L22
LOCK_COIL_2
C111
COIN_CNT_1_B
L21
COIN_CNT_1
C110
COIN_CNT_2_B
L20
COIN_CNT_2
C109
VCC
RRC1
RRC
COIN_1_B
COIN_2_B
COIN_1_F
COIN_2_F
RRC2
SERVICE_B
1P_START_B
2P_START_B
SERVICE_F
1P_START_F
2P_START_F
RRC3
TILT_B
TEST_B
1P_UP_B
1P_DOWN_B
TILT_F
TEST_F
1P_UP_F
1P_DOWN_F
RRC4
1P_LEFT_B
1P_RIGHT_B
1P_BTN_1_B
1P_BTN_2_B
1P_LEFT_F
1P_RIGHT_F
1P_BTN_1_F
1P_BTN_2_F
RRC5
1P_BTN_3_B
2P_UP_B
2P_DOWN_B
2P_LEFT_B
1P_BTN_3_F
2P_UP_F
2P_DOWN_F
2P_LEFT_F
RRC6
2P_RIGHT_B
2P_BTN_1_B
2P_BTN_2_B
2P_BTN_3_B
2P_RIGHT_F
2P_BTN_1_F
2P_BTN_2_F
2P_BTN_3_F
RRC7
2P_BTN_5_B
2P_BTN_4_B
1P_BTN_5_B
1P_BTN_4_B
2P_BTN_5_F
2P_BTN_4_F
1P_BTN_5_F
1P_BTN_4_F
{COINF}
X2
OSC-12.000MHz
IC54B
74LS04
COMRES
R26
330
MODE
IC53
TC0030CMD
RESETDTACK
CDTACK
CLK
MODE1
MODE
COM
COMINT
CS
IC80C
VBL
NMI
INT1
RW
MA[1..11]
MD[0..7]
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
D7
D6
D5
D4
D3
D2
D1
D0
ANO
AN1
AN2
AN3
AN4
AN5
AN6
AN7
{PC}
{PA}
{AN}
{MBTN}
IC54C
JP2
10k x 8
RB1
PB[2..7]
{COIN}
IC73
PC050CM
COIN2
COIN1
COINCNT1
COINCNT2
COINLOCK1
COINLOCK2
UK
VDD
+12V
IC74A
IC74B
IC74C
IC74D
74LS02
RB4
1k x 8
SWB[1..8]
SWB
SW_DIP_x08
IC78
74LS244
DIPSWB
RB3
SWA
SWA[1..8]
IC77
DIPSWA
IC79
PLRBTNS
L8
GNDA
L12
L9
L10
L11
220uF
C95
C97
C101
10nF
L13
C98
C102
L15
1000uF
C84
C100
C104
L7
SPEAKER-
JAMMA_CONN
COIN_CNT_1
COIN_CNT_2
LOCK_COIL_1
LOCK_COIL_2
SPEAKER+
SPEAKER_(+)
SPEAKER_(-)
NOT_USED
RED
BLUE
GREEN
SYNC
VIDEO_GND
SERVICE
TEST
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
PWR_FLAG
Ulf Skutnabba, twitter: @skutis77
Taito K1100319A
Sheet: /IO/
File: io.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
KiCad E.D.A. 8.0.9
Rev:
Id: 5/8

IC58 Z80CPU
IC59 YM2151
IC60 YM3012
2-CHANNEL SERIAL INPUT FLOATING D/A CONVERTER (DAC-16S)
IC67A TL074
IC67D TL074
IC67B TL074
IC67C TL074
IC68A UPC4556
IC68B UPC4556
IC67E TL074
IC68C UPC4556
IC65 MB3735
IC66 74LS138
address decoder
IC45 PC060HA
IC44 MB8464
IC43 B22-14 27512
IC31B 74LS32
IC27B 74LS08
IC27A 74LS08
Conn_01x03
Next to VR1
SD[0..7]
SA[0..15]
MD[0..3]
SPEAKER+
SPEAKER-
JAMMA11
FML
FMR
SRAM
YM2151
PC6
SRST
SNMI
SWR
SRD
CLK4M
CLK8M
SMREQ
PC060RD
PC060WR
MA1
RESET
SINT
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Taito K1100319A
Sheet: /Sound/
File: sound.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
Rev:
KiCad E.D.A. 8.0.9
Id: 6/8

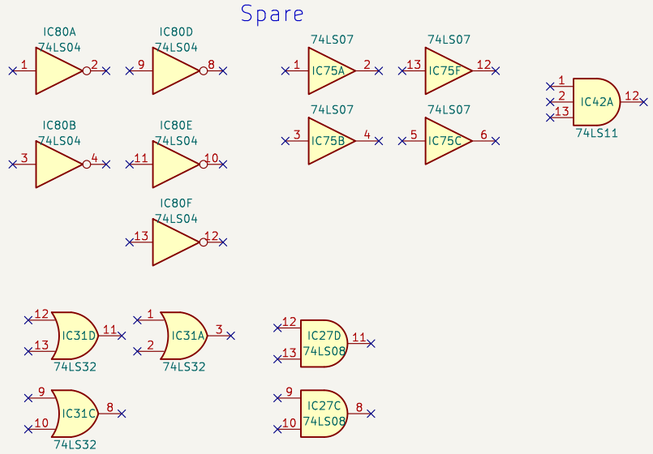
Watchdog and reset



Clocks



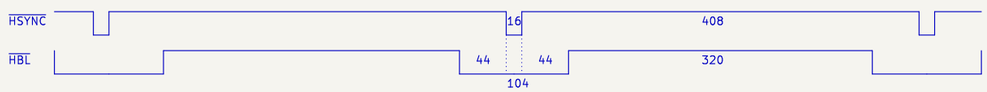
Spare



Horizontal and vertical synch timing diagrams

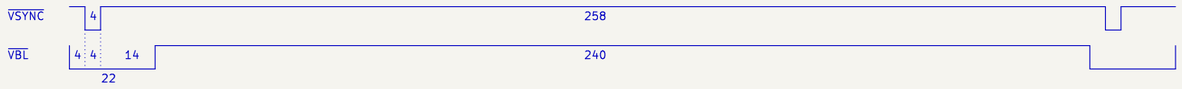
The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: f = 26.686MHz / 4 = 6.6715MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.



HSYNC and HBL
Frequency f = 6.6715MHz / 424 = 15.734670kHz.
Period T = 1/f = 63.5539us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.734670kHz / 262 = 60.05599Hz
Period T = 1/f = 1 / 15.734670kHz = 16.6511ms

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Sheet: /Misc/
File: misc.kicad_sch

Title: Rainbow Islands Extra

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